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Computer-Based Experiment for Determining Planck's Constant Using LEDs

Feng Zhou, Indiana University of Pennsylvania, Indiana, PA
Todd Cloninger, Cleveland Community College, Shelby, NC

Visible light emitting diodes (LEDs) have been widely used as power indicators. However, after the power is switched off, it takes a while for the LED to go off. Many students were fascinated by this simple demonstration. In this paper, by making use of computer-based data acquisition and modeling, we show the voltage across the LED undergoing an exponential decay after the power is switched off. We also describe a new approach for determining Planck's constant using LEDs. The simple experiment can be used either in an interactive lecture demonstration or an entry-level physics lab.

Planck's constant, $h = 6.62607554 \times 10^{-34}$ J-s, is one of the fundamental constants of nature relating to the quantum concept in modern physics. It has been previously suggested that Planck's constant be determined using LEDs,[1-5] where the key parameter to be determined was the voltage required to switch on an LED of known optical frequency. Of the two main methods reported, one uses a voltmeter to measure the minimum voltage needed to activate the LED, and the other determines "turn-on" voltage from the current-versus-voltage (I-V) curve of the LED. However, the value obtained for Planck's constant was influenced by human judgment in deciding when the LED started to emit for the first technique, and what feature of the I-V curve constituted the LED "turn-on" point for the second technique. In this paper we propose a computer-based approach that utilizes curve fitting to determine the voltage required to switch on an LED. The experiment is reliable, repeatable, and accurate.

This approach monitors only the discharge of a capacitor (C) through an LED in series with a current limiting resistor (R). Measuring the voltage across the capacitor during the discharge reveals an exponential decay that approaches a non-zero constant. That constant voltage (V_0) is the minimum voltage required for current to flow through the LED. A photon generated by the LED possesses energy $E = hf = eV_0$, where $f (= c/\lambda)$ is the frequency of the electromagnetic wave, $e = 1.6022 \times 10^{-19}$ C is the charge on an electron, and V_0 is the turn-on voltage of the LED. In terms of the wavelength λ and the speed of light c ($= 2.9979 \times 10^8$ m/s), we have:

$$h = eV_0 \frac{\lambda}{c}.$$

As shown in Fig. 1, the experimental setup consists of a circuit with a 6-V dc source, a current limiting resistor ($R = 100\ \Omega$), a capacitor ($C = 0.25$ F), and several LEDs that emit a variety of wavelengths. A 6-V dc power supply is used because the voltage probe measures a maximum voltage of 6 V. With a 100-Ω resistor, the circuit current is limited to

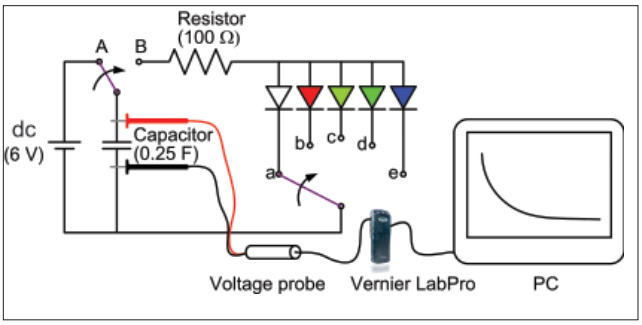


Fig. 1. Experimental setup.

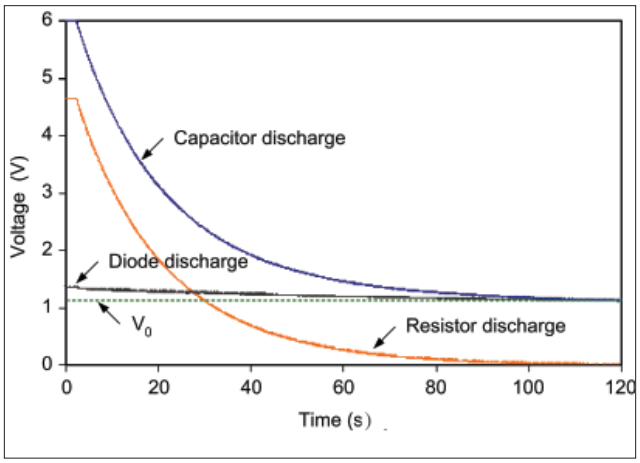


Fig. 2. Typical voltage discharge curves for the IR LED. The exponential decay curve fitting of the capacitor discharge voltage indicates it stops at a minimum voltage V_0 = 1.17 V.

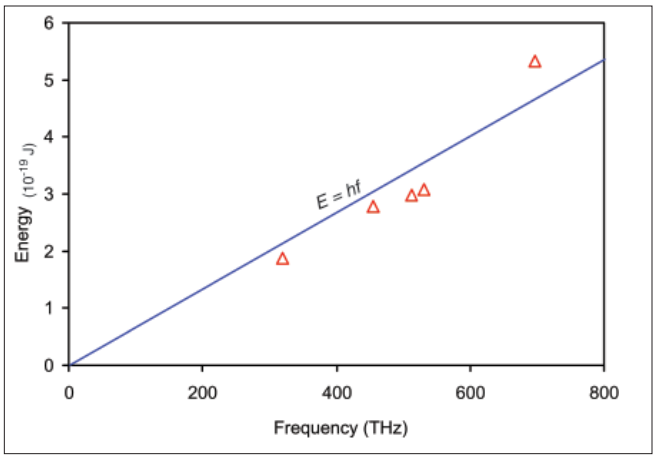


Fig. 3. Graph of photon energy $E = eV_0$ vs frequency ($f = c/\lambda$) for the five LEDs from which h is determined from the slope.

Table I. Exponential curve fitting of the capacitor discharge voltage with five different LEDs.

Light Emitting Diode	Nominal Wavelength λ [nm]	Frequency f [THz]	B (= V_0) [V]	Energy $E = eB$ [10^{-19} J]
blue	430	697.209	3.321	5.321
green	565	530.619	1.918	3.073
yellow	585	512.479	1.854	2.971
red	660	454.242	1.74	2.788
infrared	940	318.936	1.167	1.870

60 mA before an LED is inserted. A capacitor of 0.25 F in series with the resistor gives an RC time constant of 25 s, which means the exponential decay will complete within approximately 75 s. After the capacitor is fully charged by the dc source, the switch is moved from position A to position B to begin the discharge with the first LED (labeled with a). The voltage across the capacitor is recorded with a voltage probe attached to Vernier's LabPro® interface connected to a computer running Logger Pro®. Then the same procedure is repeated for other LEDs labeled with b, c, d, and e. We collected data for 5-mm blue, green, yellow, red, and infrared (IR) LEDs (from Radio Shack) with nominal emission wavelengths of 430 nm, 565 nm, 585 nm, 660 nm, and 940 nm, respectively. The actual peak wavelength could be within ± 5 nm from the nominal wavelength, according to the specification. If an additional voltage probe is used to record the voltage drop across the resistor, Kirchhoff's laws enable us to compute the LED voltage and current as well.

Figure 2 shows typical discharge curves with the IR LED used in the circuit when two voltage probes were used. Although only the capacitor voltage is required to implement our technique, the other voltages are included here for clarity. The voltage across the resistor approaches zero and is proportional to the current in the circuit while the capacitor voltage decays exponentially from its initial maximum toward a lower limit. As a result, the voltage across the LED only varies slightly approaching its minimum value as the current through the resistor approaches zero. An exponential decay function of the form $V = Ae^{-\alpha t} + B$ fits the capacitor voltage discharge curves very well, while A, B, and α are constants. Table I lists the B coefficients associated with different LEDs obtained from curve fittings, which indicate the low voltage limits, i.e., the turn-on voltages. The value of Plank's constant obtained from the graph shown in Fig. 3 is 6.625×10^{-34} J-s.

We have shown that by measuring the discharge voltage of the capacitor in the LED circuit, we can accurately determine the threshold voltage to turn on the LED by fitting the exponential decay curve. This method is reliable and consistent, generating a more accurate value than the reported methods using LEDs. The experiment exhibits desirable characteristics for an entry-level physics course or an interactive lecture demonstration while making it interesting and easy to obtain an important physical constant such as Planck's constant h, the speed of light c, or the basic

charge on an electron *e*. However, these LED-based investigations will not give high-precision measurements of the Planck's constant. Because of the particular energy bandgap structure of the LED material, the temperature effect, and the emission spectrum width, etc., the obtained *h* values vary from 5.77×10^{-34} to 7.67×10^{-34} for the five LEDs. This experiment can be further improved by using the measured emission wavelength of each LED instead of its nominal value. The simple setup can also be used to study the I-V curve and the discharge time constant of an LED.

Acknowledgment

The authors would like to thank ABPFI 2007 Workshop for providing the equipment and support to this project.

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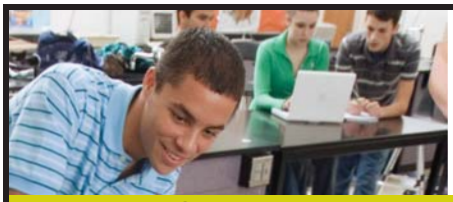
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Feng Zhou *received his PhD from Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Sciences, in 1989. Currently he is mainly involved with the electro-optics program at Indiana University of Pennsylvania. His research interests include nonlinear optics, laser R&D, and integrated photonics.*

Physics Department, Indiana University of Pennsylvania, 975 Oakland Ave., Indian, PA 15705; fzhou@iup.edu

Todd Cloninger *completed a bachelor's degree at Warren Wilson College in 1984 and an MS in applied physics at Oregon Graduate Center in 1989. His PhD research utilized near-field microscopy and porous silicon spectroscopy. Industry work involved automation and holography. Currently he teaches physics and math.*

Cleveland Community College, 137 South Post Road, Shelby, NC 28152; cloningert@clevelandcommunity college.edu



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